1. (Fall 2024 #5) The relation

$$xy^2 + ax^3 + y - a + \int_0^y e^{t^2}\,dt = 0$$

defines y as a function of x near the point $(1, 0)$ for any real number a.

(a) Find $\frac{dy}{dx}\Big|_{x=1}$. Your answer will depend on the constant a. (Hint: Consider the Fundamental Theorem of Calculus, and do not integrate.)

$$\frac{d}{dx}\left[xy^2 + ax^3 + y - a + \int_0^y e^{t^2}dt = 0\right]$$

$$y^2 + 2xy\frac{dy}{dx} + 3ax^2 + \frac{dy}{dx} + e^{y^2}\frac{dy}{dx} = 0$$

At $x=1, y=0$: $\quad 3a + \frac{dy}{dx}\Big|_{x=1} + \frac{dy}{dx}\Big|_{x=1} = 0 \quad \Rightarrow \quad \frac{dy}{dx}\Big|_{x=1} = \boxed{-\frac{3a}{2}}$

(b) For what value of a does the tangent to the graph at $(1, 0)$ pass through the point $(0, 1)$?

The tangent line to the graph at $(1,0)$ has slope $-\frac{3a}{2}$ (part (a))
The line connecting $(1,0)$ to $(0,1)$ has slope $\frac{0-1}{1-0} = -1$, so for the tangent line to pass through $(0,1)$, $-\frac{3a}{2}$ must be equal to -1.

$\Downarrow$

$$\boxed{a = \frac{2}{3}}$$

2. (Fall 2023, #3)

(a) Given the function

$$h(x) = \int_0^{\ln x} e^{2u}\,du$$

find $h'(1)$.

$$h'(x) = \frac{d}{dx}\left(\int_0^{\ln x} e^{2u}\,du\right)$$

$$= e^{2\ln x} \cdot \frac{1}{x}$$

$$h'(1) = e^{2\cdot 0} \cdot \frac{1}{1} = \boxed{1}$$

(b) Given the function

$$p(x) = x^{\sqrt{x}} = e^{\sqrt{x}\ln x}$$

find $p'(1)$.

$$p'(x) = \frac{d}{dx}\left(e^{\sqrt{x}\ln x}\right)$$

$$= e^{\sqrt{x}\ln x}\left(\frac{\ln x}{2\sqrt{x}} + \frac{\sqrt{x}}{x}\right)$$

$$= x^{\sqrt{x}}\left(\frac{\ln x}{2\sqrt{x}} + \frac{1}{\sqrt{x}}\right)$$

$$p'(1) = 1\cdot\left(\frac{0}{2} + \frac{1}{1}\right) = \boxed{1}$$

3. (Fall 2022, #3)

(a) Let

$$\frac{d}{dx}\left[\ln\left(x^2+y^2\right)=\sin(x+1)e^y\right]$$

Compute $\frac{dy}{dx}$.

$$\frac{2x+2yy'}{x^2+y^2}=\cos(x+1)\,e^y+\sin(x+1)e^y y'$$

$$\left(\frac{2y}{x^2+y^2}-\sin(x+1)e^y\right)y'=\cos(x+1)\,e^y-\frac{2x}{x^2+y^2}$$

$$\boxed{y'=\frac{\cos(x+1)\,e^y-\frac{2x}{x^2+y^2}}{\frac{2y}{x^2+y^2}-\sin(x+1)e^y}}$$

(b) Let

$$F(x)=\int_{\pi}^{x^3}\frac{\cos(t)}{t}dt$$

Compute $\frac{dF}{dx}$.

$$\frac{dF}{dx}=\frac{d}{dx}\left(\int_{\pi}^{x^3}\frac{\cos(t)}{t}dt\right)$$

$$=\frac{\cos(x^3)}{x^3}\cdot 3x^2$$

$$=\frac{3\cos(x^3)}{x}$$

(c) Find the equation of the tangent line of $F(x)$ in part (b) at $x=\sqrt[3]{\pi}$.

$$F(\sqrt[3]{\pi})=\int_{\pi}^{\pi}\frac{\cos(t)}{t}dt=0 \longrightarrow \text{point}=(\sqrt[3]{\pi},0)$$

$$F'(\sqrt[3]{\pi})=\frac{3\cos(\pi)}{\sqrt[3]{\pi}}=\frac{-3}{\sqrt[3]{\pi}} \longrightarrow \text{slope}=\frac{-3}{\sqrt[3]{\pi}}$$

the equation of the tangent line can be found using the point-slope formula:

$$y-0=\frac{-3}{\sqrt[3]{\pi}}(x-\sqrt[3]{\pi})$$

$$\boxed{y=\frac{-3}{\sqrt[3]{\pi}}x+3}$$

4. (Fall 2021, #9) Consider the following function

$$F(x) = \int_{e^x}^{\ln(x)+e} \frac{1}{\sqrt{t^2+1}}\,dt$$

(a) Calculate $F'(x)$.

$$F'(x) = \frac{d}{dx}\left(\int_{e^x}^{\ln(x)+e} \frac{1}{\sqrt{t^2+1}}\,dt\right)$$

$$= \boxed{\frac{1}{\sqrt{(\ln(x)+e)^2+1}} \cdot \frac{1}{x} - \frac{1}{\sqrt{e^{2x}+1}} \cdot e^x}$$

(b) Find the domain of $F'(x)$.

Since both F and F′ are well-defined for all $x>0$, the domain of F′ is $\boxed{(0,\infty)}$.

(c) Find the linearization of $F(x)$ at $x = 1$ and use it to estimate

$$\int_{e^{1.1}}^{\ln(1.1)+e} \frac{1}{\sqrt{t^2+1}}\,dt = F(1.1)$$

You may leave your final answer unsimplified.

$$F'(1) = \frac{1}{\sqrt{e^2+1}} \cdot 1 - \frac{1}{\sqrt{e^2+1}} \cdot e = \frac{1-e}{\sqrt{e^2+1}}$$

$$F(1) = \int_e^e \frac{1}{\sqrt{t^2+1}}\,dt = 0$$

The linear approximation at $x=1$ is $L(x) = F'(1)(x-1) + F(1)$

$$= \frac{1-e}{\sqrt{e^2+1}}(x-1) + 0$$

To estimate $F(1.1)$, we use the linear approximation

$$L(1.1) = \boxed{\frac{1-e}{\sqrt{e^2+1}} \cdot (0.1)}$$

5. (Spring 2021, #9) Consider the function

$$F(x) = \int_{\cos(x+\frac{\pi}{2})}^{\sin(x)} \frac{1}{1+t^2}\,dt$$

(a) Calculate $F'(x)$.

$$F'(x) = \frac{d}{dx}\left(\int_{\cos(x+\frac{\pi}{2})}^{\sin x} \frac{1}{1+t^2}\,dt\right)$$

$$= \boxed{\frac{\cos x}{1+\sin^2 x} - \frac{-\sin(x+\frac{\pi}{2})}{1+\cos^2(x+\frac{\pi}{2})}}$$

(b) Find the linearization of $F(x)$ at $x = 0$ and use it to estimate

$$\int_{\cos(-0.02+\frac{\pi}{2})}^{\sin(-0.02)} \frac{1}{1+t^2}\,dt \quad = F(-0.02)$$

Give your answer in decimal form.

$$F'(0) = \frac{1}{1} - \frac{-1}{1} = 2$$

$$F(0) = \int_0^0 \frac{1}{1+t^2}\,dt = 0$$

The linear approximation at $x=0$ is $L(x) = F'(0)(x-0) + F(0)$

$$= 2x$$

To estimate $F(-0.02)$, we use the linear approximation

$$L(-0.02) = \boxed{-0.04}$$